6

Indoor Air Quality (IAQ)

A separate chapter is provided on indoor air quality (IAQ) because of the importance of non-contaminated air within the fab area and surrounding support/office areas. IAQ is a subject that is continually occupying the time of industrial hygiene personnel in the semiconductor industry. Information on relevant IAQ issues are also covered in the chapter on ventilation (Ch. 4). The purpose of this chapter is to provide a general overview of IAQ terminology, odor identification, recognition and evaluation techniques, and prevention of IAQ problems. Issues that pertain specifically to semiconductor fabs or facilities will be highlighted.

1.0 RECOGNITION OF IAQ PROBLEMS

The major problems that are typically identified in the investigation of IAQ complaints can be placed into three general categories listed with decreasing frequency:[1]

- inadequate ventilation
- chemical contamination
- microbiological contamination

A brief discussion of each of these categories is appropriate.

1.1 Inadequate Ventilation

This is the largest single source of IAQ problems. Although varied, these ventilation problems often allow contaminant build-up in occupied spaces to a level that adverse health effects are experienced or the environment is uncomfortable to the employees or fab area workers.

1.2 Health Symptoms and Signs

Sick building syndrome (SBS) symptoms include: headache, dizziness, eye irritation, dry throat, nose irritation, skin irritation, sinus congestion, coughing, sneezing, shortness of breath, nausea, and fatigue. SBS symptoms are generally fairly subjective and similar to many other medical conditions. This lack of a clearly defined set of unusual symptoms significantly complicate the recognition and evaluation of IAQ incidents. Persons who wear contact lenses in SBS buildings often experience eye irritation, and may act as early warning sentinel for other less sensitive occupants of the building.

IAQ-related signs and symptoms are rarely associated with objective clinical findings. In order to be defined as SBS: the attack rate should be at least 20% of the population; there should be some commonality among the symptoms; and a clear temporal association between symptoms and building occupancy should be apparent.

Three indicators that an acute workplace exposure may be the cause of symptoms; are:

1. Symptoms disappear shortly after leaving work
2. Symptoms disappear over the weekend
3. Symptoms worsen during the work day

2.0 ODOR IDENTIFICATION

Many times odors are associated with chemical contaminants coming from inside or outside the fab manufacturing area. The complexity of the semiconductor manufacturing process coupled with tremendous volumes of air circulation by the vertical laminar flow hoods make the identification of the odor sources very difficult. Also, the use of a wide variety of toxic substances in the process adds to concern by employees in the fab area that unusual odors may indicate a potentially harmful exposure.

Often, these odors are transient in nature which makes response time a critical element in their identification.

Table 6.1 identifies some of the more common examples of transient odors that may be found in the fab area, and direct reading instruments used to determine the level of IH significance. While direct reading instruments are typically adequate for identifying odors at concentrations that are significant from a health concerns perspective, often the odor threshold for these compounds is less than the detection limit for the instrument.

Table 6.1. Transient odors and portable IH monitoring techniques.

Odors or Unknown	IH Techniques
Solvent-type (glycol-ethers or photoresists)	Combustible gas, photoionization detectors (PID), or infrared detectors
Corrosive-type (HCl, chlorine, or acids)	Detectors tubes or colormetric tape detectors
Toxic (arsine, phosphine, diborane, stibine)	Colormetric tape detectors or PID
Burning-type (decomposition or combustion products)	PID, carbon monoxide and nitrogen oxides - detector tubes or dedicated monitors
Plastic/rubber-type (wiring, gaskets, insulation, etc.)	PID, carbon monoxide and nitrogen oxides - detector tubes or dedicated monitors
Paint or glue-type (typically floor or wall painting, or sealing)	Combustible gas, PID, or detector tubes
Vinegar-type (RTV sealant, or acetic acid)	Detector tubes
HEPA filter-type (new HEPA filters)	PID or olfactory sensing
Sewer-type (sanitary sewer or acid waste lines)	Detector tubes for hydrogen sulfide or olfactory sensing
Prickly or irritating-type (acid gases)	Detector tubes or colorimetric tape detectors

Figure 6.1 is a flow diagram that outlines a systematic approach to track down the source of an odor complaint.

Odors of unknown composition associated with specific processes are a particular challenge within the semiconductor industry. One example of this is the variety of reaction products formed during plasma etching processes sometime result in strong odors that occur during maintenance. Of the plasma systems used in semiconductor manufacturing, the greatest assortment of toxic and carcinogenic compounds are present in plasma metal etching systems with chlorine-based chemistries. See Table 6.2 for further information on some plasma metal etcher compounds that may be formed during semiconductor manufacturing operations.[2] However, other plasma etching operations also have potential for significant odor generation. Metal etcher maintenance operations of greatest concern are: reaction chamber cleans, vacuum pump maintenance, and cleaning cold traps.[2]-[9]

Various parent compounds, operating characteristics, and plasma etchers are used in etching operations. This results in diverse concentrations and compounds in the breathing zone of the maintenance technician. Currently no procedure exists to determine what reaction products will form in a particular plasma etcher. Information from past plasma etcher surveys can be used to partially determine a sampling strategy for a particular etch system. However, to a certain extent, each different system must be approached as an unknown. Therefore, two separate surveys are usually necessary to evaluate a plasma etcher maintenance operation.

In the first survey, head space and bulk air samples are taken to identify possible airborne compounds from the particular operation. For reaction chamber cleans, bulk air samples are taken inside the reaction chamber. Also, a residue sample is taken from the reaction chamber for subsequent head space analysis. For cold trap surveys, air samples are taken inside the trap and at the discharge port. In pump oil surveys, a sample of used oil is taken for a head space analysis and air samples are taken above the used oil as it is transferred. Used oil filters can be evaluated by placing them in a plastic bag, allowing them to off-gas, and then analyzing the head space of the bag. Air samples should also be taken near the old oil filter as it is removed.

These samples are analyzed for certain compounds identified in previous surveys and for unknowns. Bulk air samples, coupled with *concentration mass spectroscopy* (MS/MS), can be used for the detection and identification of volatile inorganics and organics. Unknown volatile organics can also be identified using Tenax® tubes with subsequent thermal desorption and *gas chromatography/mass spectroscopy* (GC/MS) analysis.

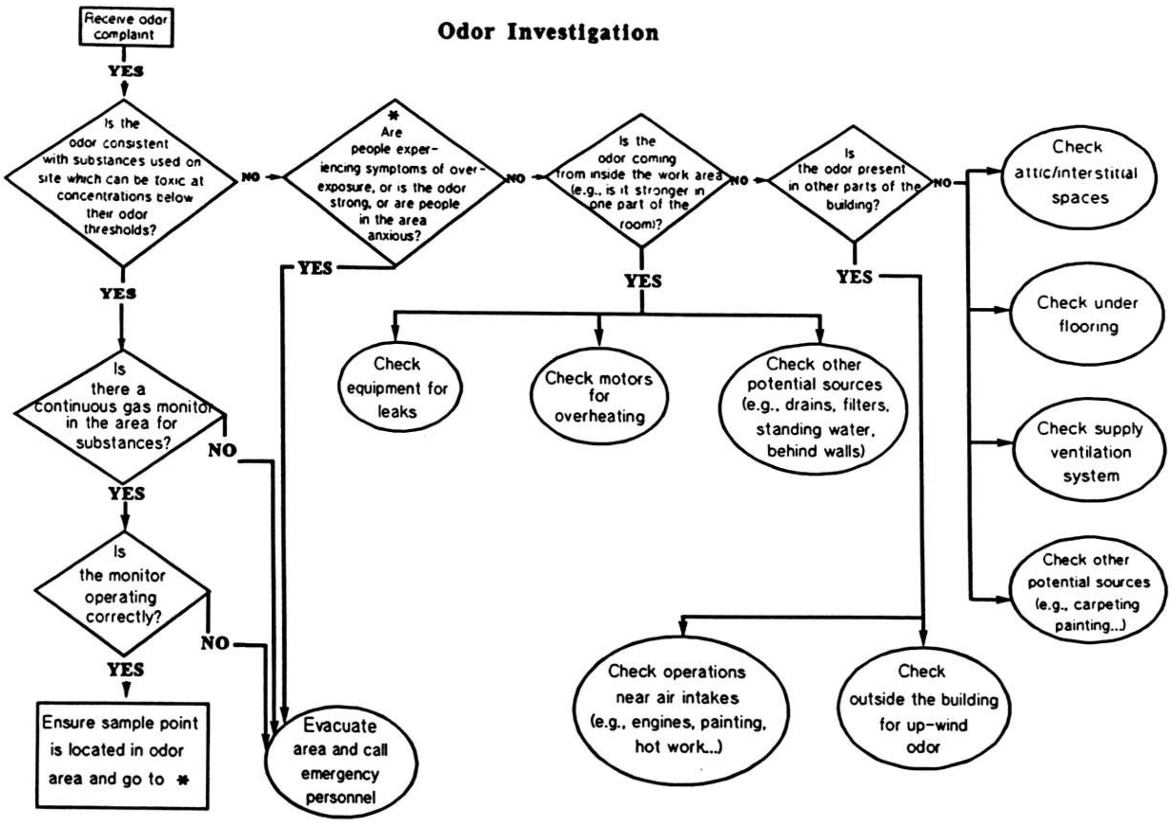

Figure 6.1. Odor investigation.

Table 6.2. Plasma Metal Etchers—Some Volatile Compounds Generated from Parent Chemistry.*[2]

Carbon Tetrachloride	Chlorine
Dichlorobenzene	Cyanogen
Hexachlorobutadiene	Cyanogen Chloride
Hexachloroethane	Hydrogen Chloride
Methylene Chloride	Hydrogen Cyanide
Pentachloroethane	Phosgene
Perchloroethylene	Toluene
Tetrachlorobutadiene	

* Parent chemistry consisting of CCl_4, Cl_2, Ar_2, and N_2 for low humidity.

The results of the first survey are used to determine the sampling strategy for a breathing zone survey. Data from the first survey can also be used in evaluating the adequacy of chemical protective clothing used by the technician.

In the second survey, personal breathing zone samples are taken using standard occupational hygiene procedures (e.g., NIOSH and OSHA methods). Data from the second survey are used to characterize the operation and to evaluate the need for airborne emission controls.

3.0 CAUSATIVE AGENTS

There are three primary categories of agents that are typically associated with an IAQ incident: physical agents, chemical agents, and microbiological agents. It is a fairly rare occurrence where one of these primary categories act as the only causative agent (e.g., a chemical spill, leak, or acute influx of the agent). A more common situation is where a synergistic combination of two of more indoor pollutants whose impacts differ from simple additive impact. The synergistic effects of many indoor pollutants can be considerably greater than the additive or cumulative effects of those pollutants. The combination of smoking and radon inhalation is one indoor air synergism that leads to a pollutant stronger than either of its constituents. Smoking and asbestos similarly are a synergistic combination, as are smoking and certain organic compounds.

3.1 Physical Agents

Temperature, humidity (or relative humidity), and indoor lighting levels are the three physical agents that impact IAQ.

3.2 Chemical Agents

Many specific indoor air pollutants have been associated historically with IAQ incidents. Table 6.3 identifies some of the specific indoor air pollutants, their possible sources, and possible health effects.[10]

Table 6.3. Some of the specific indoor air pollutants, their possible sources, and possible health effects.[10] *(Reprinted with permission from* Job Safety and Health, *pp. 715:4071–4078. © 1993 by The Bureau of National Affairs, Inc. (800-372-1033)*

Contaminant	Sources	Acute Health Effects
Acetic Acid	Etchant, silicone caulking compounds, coffee maker cleaner.	Eye, respiratory and mucous membrane irritation.
Carbon Dioxide	Improperly vented devices, free flowing nitrogen wands or blankets, human respiration, inadequate fresh air or makeup air in HVAC.	Difficulty concentrating, drowsiness, increased respiration rate.
Carbon Monoxide	Tobacco smoke, fossil fuel engine exhausts, improperly vented fossil fuel appliances.	Dizziness, headache, nausea cyanosis, cardiovascular effects, death.
Formaldehyde	Off-gassing from urea -formaldehyde foam insulation, plywood, particle board, paneling, carpeting and fabric, glues and adhesives, and combustion products including tobacco smoke.	Hypersensitive or allergic reactions; skin rashes; eye, respiratory and mucous membrane irritation; odor annoyance.

(Cont'd)

Table 6.3. (Cont'd)

Contaminant	Sources	Acute Health Effects
Nitrogen Oxides	Combustion products from gas furnaces and appliances; tobacco smoke, welding, and gas/diesel engine exhausts.	Eye, respirator, and mucous membrane irritation.
Ozone	Copy machines, laser printers, electrostatic air cleaners, electrical arcing, smog, ozonators on cooling towers, and formed from high power UV lamps in photolithography.	Eye, respiratory tract, mucous membrane irritation; aggravation of chronic respiratory diseases; contact lense wearers having extreme difficulty.
Radon	Ground beneath buildings, building materials, and groundwater.	No acute health effects are known but chronic exposure may lead to increased risk of lung cancer.
4-Phenylcyclohexene (4-PC)	Carpet backing binder made with S-B-Rubber latex.	Headaches, nausea, Upper Respiratory Irritation (URI); eye and skin irritation.
Volatile Organic Compounds (VOC's) including glycol ether derivatives, acetone, trichloroethane, benzene, toluene, xylene, methyl ethyl ketone, alcohols, methacyralates, acrolein, polycyclic aromatic hydrocarbons, and pesticides	Photolithographic chemicals, paints, glues, cleaning compounds, photocopiers, "spirit" duplicators, silicone caulking compounds, asphalt and asphaltic roofing materials, gasoline vapors, combustion products, tobacco smoke, floor cleaners and waxes.	Nausea, dizziness; eye, respiratory tract and mucous membrane irritation, headache, fatigue.

(Cont'd)

Table 6.3. (Cont'd)

Contaminant	Sources	Acute Health Effects
Miscellaneous inorganic gases including ammonia, hydrogen chloride, hydrogen fluoride, sulfur dioxide, and hydrogen sulfide.	Semiconductor process equipment, blue print equipment, window cleaners, drain cleaners, combustion products, and tobacco smoke.	Eye, respiratory tract, and mucous membrane irritation; aggravation of chronic respiratory diseases.
Tobacco Smoke is a major contributor to IAQ problems and contains several hundred substances including carbon monoxide, nitrogen oxides, formaldehyde, ammonia, benzene, tars and nicotine.	Cigarettes, cigars and pipe tobacco.	Tobacco smoke can irritate the respiratory system in an allergic or asthmatic persons often results in eye and nasal irritation, coughing, wheezing, sneezing, headache, and related sinus problems. People who wear contact lenses often complain of burning, itching, and tearing eyes when exposed to tobacco smoke.
Asbestos	Insulation and other building materials such as floor tiles, dry wall compounds, reinforced plasters, ventilation system muds, and boiler/ducting coverings.	Asbestos is not normally a source of acute health effects. Chronic exposure to airborne asbestos fibers is a human carcinogen.
Man-Made Mineral Fibers	Fibrous glass and mineral wool.	Irritation to the eyes, skin, and lungs; dermatitis.

3.3 Microbiological Agents

Microbial pollutants in buildings can cause two general types of disorders: allergic and infective. Because of the wide variety of bacteria, viruses, molds, and fungi that can flourish in an indoor environment, it generally is difficult to establish exact cause and effect relationships between microorganisms in the air and disease.[11] The variation in individuals' sensitivity to these contaminants also makes it difficult to measure the extent of microbial pollutant-induced allergies. Table 6.4 identifies some microbial indoor air pollutants, their possible sources, and possible health effects.[10]

As can be seen from this general overview of the three major categories of causative agents in IAQ incidents, the contaminants are varied, the sources are diverse, and the acute health effects common to many routine sicknesses such as colds, flu, and allergic-type responses. Table 6.5 provides a matrix of the major causes of SBS and the typical symptoms associated with each of the causes.[12]

Table 6.4. Microbial indoor air pollutants, their possible sources, and possible health effects.[10] *(Reprinted with permission from* Job Safety and Health, *pp. 715:4071–4078. © 1993 by The Bureau of National Affairs, Inc. (800-372-1033)*

Contaminant	Sources	Acute Health Effects
Microoganisms and other biological contaminants include viruses, fungi, mold, bacteria, nematodes, amoeba, pollen, dander and mites.	Air handling system condensate, cooling towers, water damaged materials, high humidity indoor areas, damp organic material and porous wet surfaces, humidifiers, hot water systems, outdoor excavations, plants, animals and insects, food and food products.	Hypersensitivity diseases (hypersensitivity pneumonitis, humidifier fever, allergic rhinitis, etc.), and infections such as legionellosis are seen. Symptoms include chills, fever, muscle ache, chest tightness, headache, cough, sore throat, diarrhea and nausea.

Table 6.5. A matrix of the major causes of SBS and the typical symptoms associated with each of the causes.[12]

Symptom\Cause	Vent	VOC	Temp	Humi	CO	HCHO	O_3
Eye Irritation	X	X		X		X	X
Nose Irritation		X		X		X	X
Dry Throat	X	X		X		X	X
Skin Irritation			X				
Sinus Congestion	X					X	
Cough							X
Sneezing							
Shortness of Breath					X		
Headache	X	X	X		X		X
Nausea						X	
Dizziness							
Fatigue	X	X	X		X		X

KEY: Vent = Inadequate Fresh Air
VOC = Elevated Volatile Organic Compounds
Temp = Too Warm
Humi = Low Humidity
CO = Elevated Carbon Monoxide Concentration
HCHO = Elevated Formaldehyde Concentration
O_3 = Elevated Ozone Concentration

4.0 EVALUATION OF IAQ PROBLEMS

4.1 Major IAQ Problems

In the IAQ investigations completed by NIOSH, the sources identified for IAQ problems were found to result from:

- Inadequate ventilation in 52% of the cases
- Contamination from inside the building in 16%
- Contaminants brought in from outside the building in 10%
- Microbiological contaminants in 5%

- Building material contamination in 4% of the cases
- The remaining 13% represent those investigations where no problem could be identified[1]

4.2 Inadequate Ventilation

ASHRAE standards are generally accepted by U.S. government agencies and industry as the basis for evaluating building ventilation. ASHRAE standard 62-1989, *Ventilation for Acceptable Indoor Air Quality,*[13] and ASHRAE standard 55-1992, *Thermal Environmental Conditions for Human Occupancy*[14] are both used for this purpose. Ventilation problems commonly encountered are:

- Not enough fresh outdoor air supplied to the occupied spaces
- Poor air distribution and mixing which causes stratification, draftiness, and pressure differences between office spaces
- Temperature and humidity extremes or fluctuations (sometimes caused by poor air distribution or faulty thermostats)
- Air filtration; problems caused by improper or a complete lack of maintenance to the building ventilation system

Figure 6.2 provides a flow diagram for developing an IAQ profile of any building, fab area, or other area where IAQ has become a concern.[15]

In many cases, ventilation problems are created or enhanced by certain energy conservation measures applied in the operation of the building ventilation. These measures include reducing or eliminating fresh outdoor air, reducing infiltration and exfiltration; lowering thermostats or economizer cycles in winter, raising them in summer; eliminating humidification or dehumidification systems; and early afternoon shut-down and late morning start-up of ventilation systems.

4.3 Contamination from Inside the Building

Interior air pollution has many sources, including materials used in building construction and in building furnishings; heating, ventilation, and air conditioning systems (HVAC); office equipment; cleaning solutions and paints; and materials and substances temporarily brought into buildings. Examples of this type of problem include:

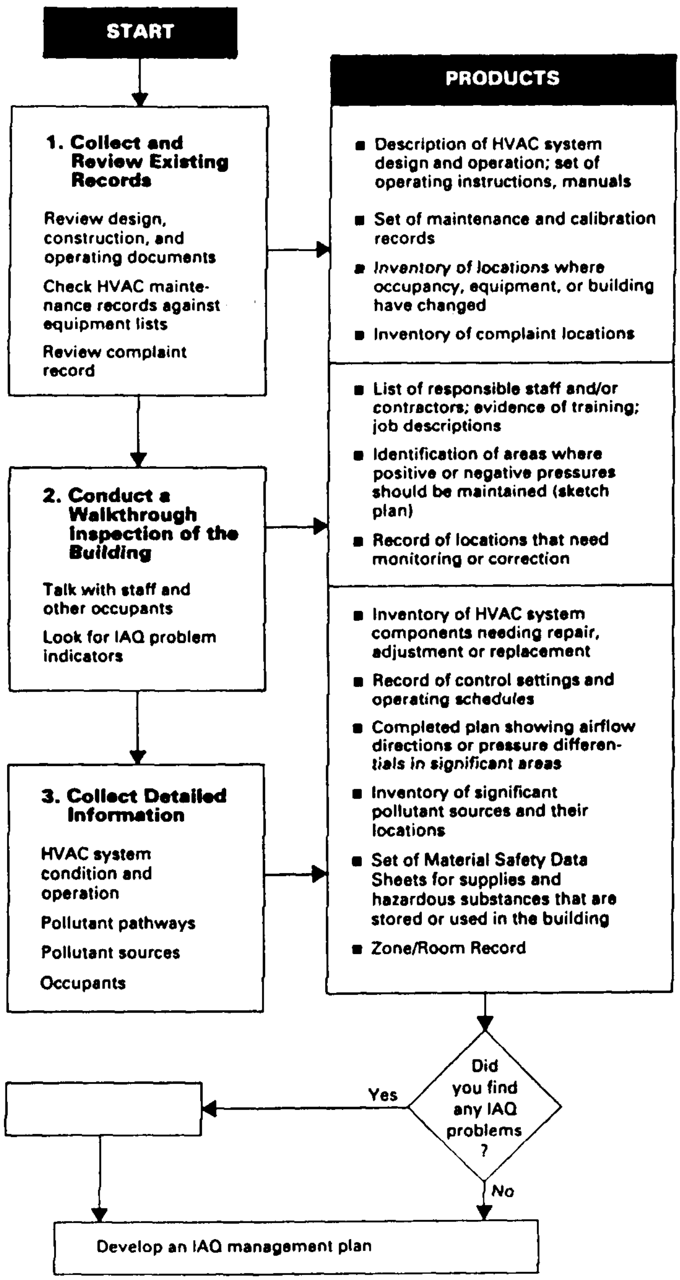

Figure 6.2. Flow Diagram for developing an IAQ Profile.[15]

- Dry ink copying machines.
- Wet copying machines:
  - Methyl alcohol from spirit duplicators.
  - Ammonia and acetic acid from blueprint copiers and silicone caulking compounds.
- Pesticides improperly applied.
- Cleaning agents such as rug shampoo improperly diluted (e.g., sodium lauryl sulfate).
- Tobacco smoke of all types (direct, side-stream, or exhaled).
- Combustion gases from sources common to cafeterias and laboratories.
- Cross-contamination from poorly ventilated sources that leak into other air handling zones.
- Food products from microwave ovens.
- Sewer gas from dry p-traps. This is a relatively common occurrence in the semiconductor industry due to evaporation of the water that is needed in the p-trap to stop sewer gases from back-flowing through floor drains, sink drains, and eyewash/deluge shower drains.

4.4 Contamination from Outside the Building

Contamination sources from outside the building are a common cause of IAQ problems in semiconductor fab areas and support buildings. They typically include:

- Problems due to motor vehicle exhaust. As mentioned in the chapter on ventilation (Ch. 4), this has been a cause of odors at many semiconductor facilities depending on the location of delivery roads, dock facilities, and prevailing wind conditions.
- Boiler gases. Odor and re-entrainment into general HVAC systems has been a source of concern in semiconductor manufacturing buildings.
- Cafeteria odors. The grilling of garlic or onions on the cafeteria grill or stove that is adjacent to a fab area have caused many precautionary evacuations (due to the similarity of arsine odor to garlic-like odors).

- Previously exhausted air. As mentioned in the chapter on ventilation (Ch. 4), the re-entrainment of LEV system effluents from wet scrubbers or exhaust conditioning systems has occasionally occurred in the semiconductor industry.
- Contaminants from construction or renovation projects such as asphalt fumes, solvent and paint vapors, welding or cutting smokes, fumes and dusts. This again has occasionally occurred in the semiconductor industry (as well as all other manufacturing operations where production must be continued while construction/modifications are made).
- Wood smoke.
- Agricultural activities such as field burnings, pesticide spraying, livestock odors, and garlic/onion harvests. This odor is especially significant when it gets pulled into fab area general HVAC systems, as the characteristic odor of arsine is garlic-like. Many fab areas in California's Silicon Valley have evacuated during the summers garlic harvesting activities from the adjacent city of Gilroy.

The first four contamination sources listed above are essentially caused by re-entrainment of outside air. They are normally the result of improperly located exhaust and intake vents or stacks, periodic changes in wind conditions, or the effects of roof screens around the perimeter of high tech buildings. These architectural facades that hide the general HVAC equipment located on the roofs cause eddy currents at roof level, and disturb air flow patterns which result in LEV exhaust streams being brought back down to roof level and re-entrained into the HVAC intakes (see Ch. 4 for additional information).

4.5 Microbiological Contamination

Microbiological contamination in buildings is often a function of moisture incursion from sources such as stagnant water in HVAC air distribution systems and cooling towers. Elimination of standing water and water treatment are the best controls for preventing microbiological contamination. See Ch 2, Sec. 3.4: Biological Hazards, for additional information.

4.6 Contamination from Building Materials and Products

Basic construction characteristics, furnishings, management practices, building population habits, and even human breathing can cause an array of pollutants and deteriorate IAQ.

Materials used in the construction of the building can become indoor air pollutants. Materials such as asbestos become airborne pollutants when they start to disintegrate due to age, thermal stress, or ruptures caused by building repairs or renovations.

Organic solvents such as formaldehyde, xylene, toluene, acetone, alcohols, and other chemicals used in wall and floor construction, and furnishings also contribute to IAQ problems over a long period of time through a slow process known as "outgassing" which is just continual evaporation. Formaldehyde can come from urea formaldehyde insulation, rugs, particle board furniture, plywood floors, and even tobacco smoke.

Examples of this type of problem include:

- Formaldehyde off-gassing from:
  - Urea-formaldehyde foam insulation
  - Particle board
  - Plywood
  - Some glues and adhesives used during construction releasing volatile organic compounds (VOC's)
- Dermatitis resulting from fibrous glass erosion in lined ventilating ducts
- Various organic solvents from glues and adhesives
- Paints and lacquers
- Acetic acid used as a curing agent in silicone caulking
- Particles from deteriorated filters

4.7 Psychosocial Considerations

Employees may report common symptoms that could be caused by numerous non-work related illnesses. In other cases, symptoms reported may be so unusual as to appear bizarre. However, no matter what the complaints or symptoms, they should be received in a professional manner, conveying concern and acceptance of their genuineness.[16]

The initial goal of an IAQ investigation is to approach the complainant with an open mind. The person's concerns must be legitimized so that he or she feels that his or her sense of distress is being taken seriously. Dismissal of the problem as insignificant or trivial will likely lead to feelings of anger and rejection. Also, uncertainties associated with the causes and symptoms of SBS make it difficult to ignore or minimize an employee's complaints, particularly in the first stages of an IAQ investigation.

5.0 GENERALLY RECOGNIZED ACCEPTABLE LEVELS

5.1 Temperature

ASHRAE standards recommend that winter air temperatures be maintained between 68–76°F (20–24°C) , and summer temperatures be maintained between 73–79°F (23–26°C). However, if the temperature is not 72°F ± 2° (22°C ± 1°), complaints may arise.

5.2 Relative Humidity

If the relative humidity in occupied spaces is greater than 60%, the growth of allergenic or pathogenic organisms will be stimulated. If the humidity in occupied spaces and low velocity ducts and plenums exceeds 70%, fungal contamination; (e.g., mold, mildew) can occur. Humidity levels below 20% are associated with increased discomfort due to electrical static discharges and drying of the mucous membranes. In typical fab clean rooms, humidity is controlled within a range (e.g., 35–55% relative humidity) for process reasons (e.g., to maintain a stable sensitivity in resists).

5.3 Carbon Dioxide

ASHRAE recommends carbon dioxide (CO_2) levels not exceed 1000 ppm.[13] The CO_2 concentration is widely used as an indicator of indoor air quality. Comfort (odor) criteria are unlikely to be satisfied if the ventilation rate is set so that 1000 ppm CO_2 is not exceeded. Some studies suggest that CO_2 levels of 600–700 ppm are necessary to maintain employee comfort.[17][18]

5.4 Supply Volumes

The general HVAC system should be evaluated to ensure it is delivering at least 20 CFM (10 L/s) per person of outside air throughout the occupied zone at all times the building is in use.[13]

5.5 Make-up Air

For new construction design, minimum outside air ventilation should be at least 20 CFM per person for office areas.[12]

5.6 Poor Mixing

Poor mixing is another potential ventilation problem sometimes found when air is delivered and returned through ceiling diffusers. Standard smoke tubes, temperature, and CO_2 measurements at one foot and seven foot heights above the floor may be useful techniques in evaluating stratification resulting from poor mixing.

6.0 INDUSTRIAL HYGIENE IAQ MONITORING EQUIPMENT

The following list of equipment is typical of the type of IH equipment that would be used in an IAQ investigation:

- Colorimetric Detector Tubes (CO_2, CO, Air Current, Total Hydrocarbons, and Formaldehyde)
- Combustible Gas and Oxygen Meter
- Continuous Gas Monitoring Devices—for carbon dioxide and carbon monoxide
- Velometer, Thermal Anemometer
- Psychrometer—handheld and strip chart
- Thermometer—handheld and strip chart
- Diffusion Detector Tubes (Dosimeter Tubes and Passive Personal Monitors)
- Oil of Wintergreen or Sulfur Hexafluoride (SF_6) (Tracer Gas)
- Smoke Bottle and Smoke Tubes
- Collection Bags and Swabs (to hold/collect settled particulate and wipe samples)

- Particle Counter—direct reading
- Air Sampling Pumps (low/high flow)
- Infrared Gas Spectrometer (e.g., Miran 1B) or electron-capture gas chromatograph—for performing tracer gas studies
- Dry ice, liquid nitrogen, or vapor wand (see Ch. 4: Ventilation)

7.0 PREVENTION OF IAQ PROBLEMS

There are two basic approaches to reducing or eliminating indoor air quality problems: treatment and prevention. The old adage of "an ounce of prevention is worth a pound of cure" could never be more applicable than to IAQ. The prevention approach attempts to eliminate pollutants at their source. The treatment approach accepts pollutants as inevitable components of indoor air and attempts to disperse them as much as possible. Solving indoor air quality problems will require both approaches, with prevention typically being the most cost effective.

7.1 Design and Construction Considerations

There are four main topics discussed in this brief overview of design and construction considerations as they relate to IAQ: building ventilation; exhaust outlets, stacks, and inlets; building bake-outs; and new materials and equipment considerations. Some basic design considerations for each of these are discussed below.

Building Ventilation. Fan systems with full economizers are recommended when they are consistent with the ownership/leasing requirements of the building. HVAC system should be designed to provide ventilation rates of 20 CFM (10 L/s) per person of outside air throughout the occupied zone at all times the building is in use.[13] It is important to ensure ventilation rates are measured when economizers are at the setting providing the minimum amount of outside air. Restrooms should have ventilation rates of 50 CFM (25 L/s) per toilet or urinal and should be at a negative pressure with respect to the rest of the building. [13] Air velocity should be 30 to 50 feet per minute at each work station when practical.[13] All buildings should be non-smoking or have smoking allowed only in a separate space vented directly to the outside. Fiberglass should not be used to line the inside of HVAC ducts. Where soils contain high concentrations of radon, ventilation practices that place crawlspaces, basements, or under-ground ductwork below atmospheric pressure will tend to increase radon concentrations in buildings and should be avoided.

Exhaust Outlets, Stacks, and Inlets. Air intakes should be located on the lower one-third of a building. The exhaust should be on the upper two-thirds of the building or on top of the building. Intakes should not be placed near loading docks, parking garages, busy streets, or other potential sources of contaminants. Exhaust outlets and stacks should be placed on the predominant downwind side of the building and intakes on the upwind side. Place exhaust stacks as far away from intakes as possible—50 feet is considered adequate in most cases. When exhaust outlets are located on the roof, aesthetic enclosures should be avoided. If required by local code, they should be of the open-louvered type which allow horizontal winds to flush the enclosure. Intakes should not be located within the enclosure. Avoid the use of rain caps which direct the flow of exhaust air back towards the roof. These caps can greatly reduce the dilution of exhausted air. Provide ample stack height. The Uniform Building Code suggests that stacks should be at least 10 feet away and 2 feet above an air intake. A guideline states that stacks within 50 feet of the roofline or an air intake should be 10 feet tall. In no cases should stacks be less than seven feet tall to protect maintenance people working near the stacks. Place cooling towers at least 25 feet from intakes.[19]

Building Bake-Outs. Currently, data supporting the effectiveness of building bake-outs is weak at best. Building bake-outs or flushing of the buildings as part of the start-up for new buildings appear to be partially effective in driving off or hastening the evaporation of volatile compounds. The issue of building bake-outs and flushing to remove chemical contaminants from new construction and renovation is still under investigation by NIOSH, and should be done with caution to avoid the delamination of glues, sealants, etc. Typically, temperatures above 85 to 90°F are considered the threshold for delamination to begin. Talk to the furniture or equipment manufacturers to be sure.[20]

New Material and Equipment Considerations. Materials low in pollutant emissions should be used whenever possible. One recent voluntary guideline for emmission criteria proposed by the Carpet and Rug Institute[21] is:

	Emission Factor $mg/m^2 \cdot hr$
Total volitile organic compounds (TVOC)	0.6
Styrene	0.4
4-Phenylcyclohexene (4-PC)	0.1
Formaldehyde	0.05

To lessen the chances of high concentrations of VOC building up, the following installation procedures should be considered for interior finishing and construction materials:

- Acquire representative samples of prospective finishing materials.
- Store samples in a closed jar and determine if odors are unacceptable by "sniff" test or by laboratory analysis.
- Review material safety data sheets for materials that vendors may use when installing finishing materials, especially adhesives or solvents.
- Require vendors to store unpackaged finishing materials in a ventilated warehouse. This is particularly important for items with large surface areas that are marginal with respect to the "sniff" test (e.g., new carpets or new partitions).
- Require finishing materials to be installed while providing maximum outdoor air ventilation. Restrict the quantities of adhesives and solvents used during installation to that which is essential. Avoid the use of chemicals which are toxic, carcinogenic, or reproductive hazards.

Contaminants from stationary local sources within the building should be controlled by collection and removal as close to the source as practical. New copy or blueprint rooms should be exhausted to the outside and be kept negative with respect to the surrounding areas.

8.0 RECORDS

All preventive maintenance documentation should be retained by the site for current year plus five years to allow verification of the design and routine operation of all building systems that can impact the indoor air.

All health and safety records pertaining to an IAQ investigation should be retained. This should include:

- All air sampling data
- All questionnaires that are returned by those involved, including a detailed listing of signs and symptoms described by affected individuals

- All data on the building system measurements made including: temperature, relative humidity, air flow, and percent outside air
- Any conclusions regarding cause and a complete list of all recommendations made
- Documentation of follow-up of the recommendations

All medical records pertaining to an IAQ problem, including initial report, clinical and/or lab findings, as well as the diagnosis by the attending physician or occupational health nurse should be retained as an exposure record. See Ch 8: IH Recordkeeping, for additional information.

Federal OSHA's proposed regulations on indoor air quality published in the Federal Register (59FR15,968) on April 5, 1994 will require various elements:

- Developing an indoor air quality compliance program and worker training program
- Retain maintenance records
- implement controls to maintain IAQ
- Either prohibit smoking in the workplace or permit it only in enclosed negative pressure area exhausted directly to the outside.

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